

Database Project Report

Course: INT1313 – Database Systems

Project Title: Freelance Marketplace Ecosystem (Mini-Upwork)

Due Date: [Insert Date – Week 3]

A. Project Identity

- **Team Name:** [G3]
- **Team Members:**
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- **Project Title:** Freelance Marketplace Ecosystem (Mini-Upwork)

1. Introduction & Project Scope

1.1 System Objective

The purpose of this project is to design and implement a relational database for a **Freelance Marketplace Ecosystem**, inspired by platforms such as Upwork.

The system connects two main types of users:

- **Clients**, who create and publish projects.
- **Freelancers**, who search for projects and submit proposals.

The database supports the complete basic workflow of a freelance marketplace. Clients can create projects, freelancers can submit proposals, and clients can select a proposal and establish a contract with a freelancer. After the work is completed, the system records payments and allows both parties to leave reviews.

The main objective is to create a well-structured relational database that ensures:

- Data consistency.
- Referential integrity.
- Reduced data redundancy.
- Secure access control.
- Efficient data retrieval.

- Support for common business operations.

The project focuses on the **database layer** rather than developing a complete web or mobile application.

1.2 Functional Requirements

The system shall provide the following functional capabilities.

User Management

The system shall allow users to:

- Register an account.
- Store personal information such as name, email, and password.
- Have one or more roles, such as Client, Freelancer, or Administrator.
- Update profile information.

Freelancer Management

The system shall allow freelancers to:

- Create a freelancer profile.
- Write a professional title and description.
- Specify their hourly rate.
- Add multiple skills.
- Associate a skill with a proficiency level.

Client Management

The system shall allow clients to:

- Create a client profile.
- Provide company or organization information.
- Post freelance projects.

Project Management

Clients shall be able to:

- Create a project.
- Define the project title and description.
- Specify a budget.
- Define the project status.

- Publish or close a project.

Freelancers shall be able to:

- Browse available projects.
- Search for projects.
- Submit proposals.

Proposal Management

The system shall allow freelancers to:

- Submit a proposal for a project.
- Specify a cover letter.
- Suggest a bid amount.
- Suggest an estimated completion period.

Clients shall be able to:

- View proposals submitted for their projects.
- Accept or reject proposals.

Contract Management

When a proposal is accepted:

- A contract can be created between the client and freelancer.
- The contract stores the agreed amount.
- The contract records start and end dates.
- The contract status can be updated.

Payment Management

The system shall:

- Record payments associated with contracts.
- Store payment amount and payment date.
- Track payment status.

Review and Rating Management

After a contract is completed:

- Clients can review freelancers.
- Freelancers can review clients.
- Each review contains a rating and optional comment.

1.3 Non-Functional Requirements and Constraints

The database must satisfy the following non-functional requirements:

Security

- User passwords must not be stored in plain text.
- Passwords should be stored as secure password hashes.
- Access to sensitive data should be controlled using roles and privileges.
- Unauthorized users must not be able to modify financial or contract information.

Data Integrity

- Primary keys must uniquely identify records.
- Foreign keys must maintain relationships between tables.
- Invalid references must not be allowed.
- Important attributes must use NOT NULL constraints where appropriate.
- Business rules should be enforced using constraints and triggers.

Performance

- Frequently searched columns should be indexed.
- Queries involving projects, proposals, and contracts should be optimized.
- The database design should support future growth.

Availability and Maintainability

The database structure should be clear, normalized, and easy to maintain. SQL code should follow a consistent style guide.